Array in C

Array

Why do we need Array?

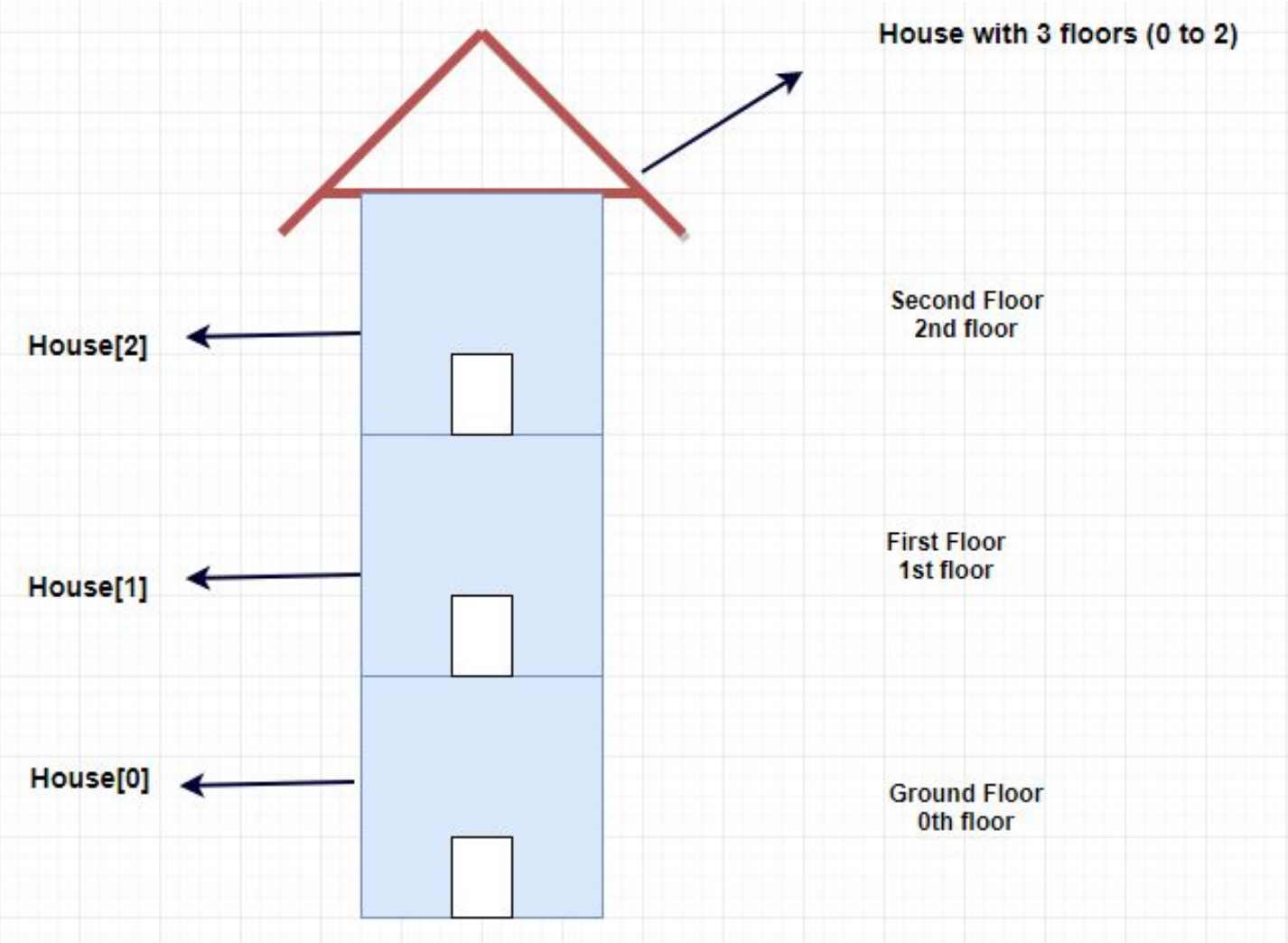
- Let's consider the situation where we need to get 10 student's age and store it for some calculation.
- Since age is an integer type, we can store it something like below,

```
Example:

int age1, age2, age3, age4, age5, age6, age7, age8, age9, age10;
```

- if we declare like this, it will be very difficult for us to manipulate the data.
- If more number of student joins, then it is very difficult to declare a lot of variables and keep track of it.
- To overcome this kind of situation, we should use **Array data structure**.

Example

- If we consider house as a datatype then apartment is an array of house.



These apartment has 3 house. i.e

House[0] -> first house

House[1] -> second house

House[2] ->third house.

Likewise, we can use array to solve our problem.

How to Declare an Array

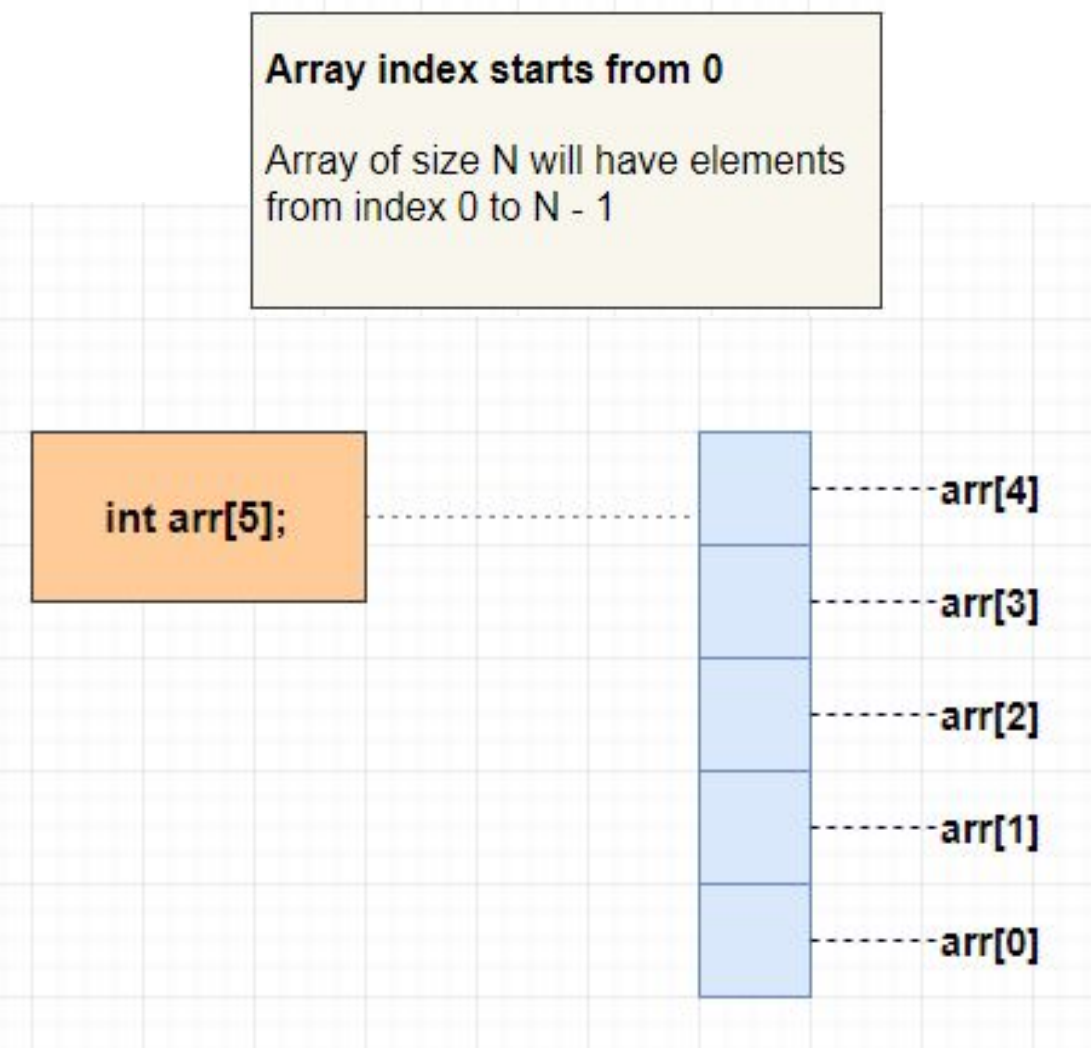
Syntax: `datatype arrayname[arraysize];`

Example:

```
int num[5];
int arr[5];
```

- `int` specifies that array will store integers
- `num` and arr is the name of the array
- `[5]` indicate that the array can hold 5 elements
- Size of the array cannot change once declared.
- `datatype` can be int, float, double, char etc.
- `arrayname` should be a valid variable name.
- `arraysize` is an integer value.

Note: arraysize should not be a negative value.



Initializing and Accessing Array Elements

```
int num[5] = {10, 20 , 30 , 40, 50};
```

- **Indexing: Array indices start from 0**

```
Here, num[0]  =  10
      num[1]  =  20
      num[2]  =  30
      num[3]  =  40
      num[4]  =  50
```

- **Access the value**

```
printf("%d", num[3]);   // print 40
```

Initializing and Accessing Array Elements

- If the initializer is shorter than the array, the remaining elements of the array are given the value 0:

  ```
  int a[10] = { 1, 2, 3, 4, 5, 6};
  ```

  in this case initial value of **a[10]** = { 1, 2, 3, 4, 5, 6, 0, 0, 0, 0};

- Initialize an array to all zeros

  int a[10] = {0};

  in this case, initial value of **a[10]** = {0, 0, 0 ,0 , 0, 0, 0, 0 , 0, 0};

- Using **for** loop

  ```
  for ( int i=0; i< 10 ; i++)
  {
    a[i] = 0;
  }
  ```

Program 1

Write a program to create an integer array named num with a size of 10, containing the elements {3, 5, 6, 9, 8, 10, 100, -3, 0, 1}, and then print all its values.

The output should be like

```
Array elements are:

num[0]  = 3     num[5]  = 10
num[1]  = 5     num[6]  = 100
num[2]  = 6     num[7]  = -3
num[3]  = 9     num[8]  = 0
num[4]  = 8     num[9]  = 1
```

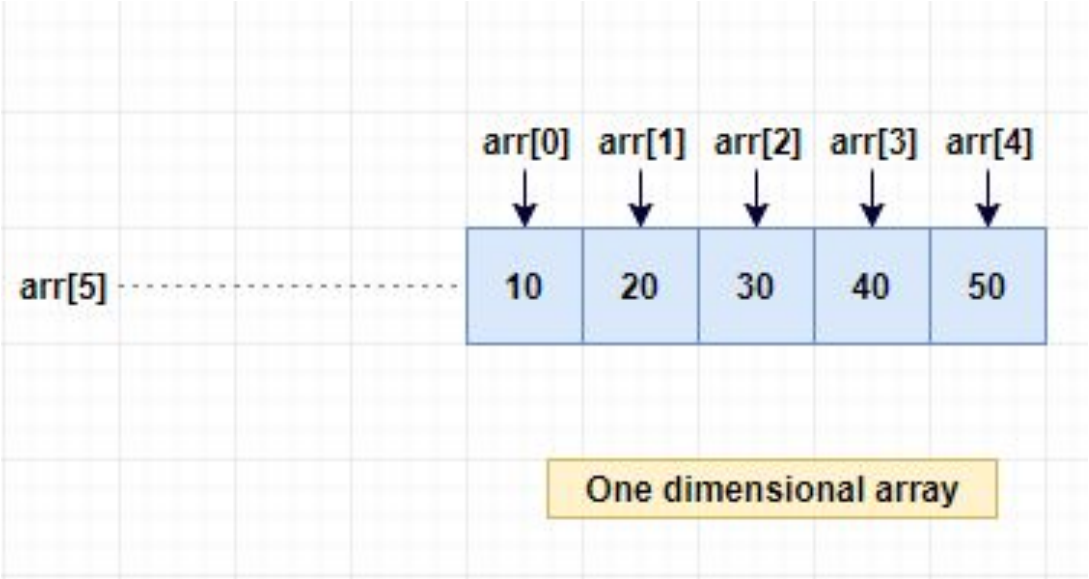
Program 1

```c
#include <stdio.h>

int main()
{
    int num[10] = {3, 5, 6, 9, 8, 10, 100, -3, 0, 1};

    printf("Array elements are:\n");
    for(int i = 0; i < 10; i++)
    {
        printf("num[%d]=%d\n", i,num[i]);
    }

    return 0;
}
```

Initialization of an Array

- We can also let the compiler to determine the array size by not giving the initial array size while initializing it.
- In this case, array size will be **determined by compiler.**

```
Example:

int arr[] = {10,20,30,40,50};
```



- **Program 2**. Write a program that prompts the user for input values, stores these values in an array using a loop, and then displays the array's contents.

Getting Input from user

- In the case of array, we know exact array size while initializing it, so we should use for loop.

```
#include<stdio.h>

int main()
{
    int arr[5];    //array of lengeth 5 type integer.. arr[0]-->arr[4]
    int i;

    for(i = 0; i < 5; i++)
    {
       scanf("%d",&arr[i]);  //for getting 5 elements from user
    }
    for(i = 0; i < 5; i++)    //printing all 5 elements
    {
      printf("%d\n",arr[i]);
    }
    return 0;
}
```

- It will get 5 elements from the user and print it.

Program

```
Consider  a[5] = { 3, 5, -2, 10, 5}
          b[5] = {-11, 3, 6, 0, 1}
```

- Write a program to find the sum of two integer array and display the result.

  The output of the program should be:

```
a[0] + b[0] = -8
a[1] + b[1] = 8
a[2] + b[2] = 4
a[3] + b[3] = 10
a[4] + b[4] = 6
```

Sum of two array

```c
#include <stdio.h>

int main()
{
    int a[5] = {3, 5, -2, 10, 5};
    int b[5] = {-11, 3, 6, 0, 1};
    int sum[5];

    // Calculate the sum of corresponding elements and store in array `sum`
    for (int i = 0; i < 5; i++)
    {
        sum[i] = a[i] + b[i];
    }

    // Display the result
    for (int i = 0; i < 5; i++)
    {
        printf("a[%d] + b[%d] = %d\n", i, i, sum[i]);
    }

    return 0;
}
```

Array Sum Calculation

Write a program that takes an integer array of size 10, containing values entered by the user. Calculate and print the sum of all elements in the array.

Sample Output:

```
Enter the value of a[0] : 3
Enter the value of a[1] : 4
Enter the value of a[2] : 5
Enter the value of a[3] : 6
Enter the value of a[4] : 7
Enter the value of a[5] : 8
Enter the value of a[6] : 9
Enter the value of a[7] : 10
Enter the value of a[8] : 11
Enter the value of a[9] : 12
```

sum = 75

Array Sum Calculation

```c
#include<stdio.h>

int main()
{
    int a[10];
    int sum = 0;

    for (int i=0; i<10; i++)
    {
        printf("Enter the value of a[%d] :", i);
        scanf("%d", &a[i]);
        sum += a[i];
    }
    printf("sum=%d\n", sum);
    return 0;
}
```

Dot product

Consider two array **num1** and **num2** of size 5, containing values entered by the user.

Find the dot product of **num1** and **num2.**

Sample output:

```
Enter the value of num1[0]: 2
Enter the value of num2[0]: 4
Enter the value of num1[1]: 5
Enter the value of num2[1]: 6
Enter the value of num1[2]: 8
Enter the value of num2[2]: 9
Enter the value of num1[3]: 2
Enter the value of num2[3]: 0
Enter the value of num1[4]: 2
Enter the value of num2[4]: 2
The value of dot product is 114
```

Dot product

```c
#include<stdio.h>

int main()
{
   int num1[5], num2[5];
   int dot_prod = 0;

   for (int i =0; i<5 ; i++)
   {
       printf("Enter the value of num1[%d]: ", i);
       scanf("%d", &num1[i]);
       printf("Enter the value of num2[%d]: ", i);
       scanf("%d", &num2[i]);
       dot_prod += num1[i]*num2[i];

   }
   printf(" The value of dot product is %d\n", dot_prod);
   return 0;
}
```